Joseph E. Velasquez

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Education

California State University Fullerton — **Fullerton, CA**
B.S. Computer Science — *Aug 2025*

Riverside City College — **Riverside, CA**
A.S. Computer Science — *May 2022*

Experience

Inland Neurosurgery Institute - Software and Systems Engineer
Aug 2025 – Oct 2025

- Supported integration and deployment of a new Electronic Medical Records (EMR) system, ensuring minimal disruption to clinical operations
- Resolved hardware, software, and network issues during EMR migration to ensure continuity of clinical operations
- Built and maintained automation workflows using Microsoft Power Automate, reducing manual administrative workload and increasing efficiency
- Assisted in maintaining cybersecurity measures, including system monitoring, patching, and basic threat mitigation
- Developed a patient scheduling chatbot using local LLM, enabling automated appointment booking and reducing administrative workload

Residential IT Support Technician (ResNet) - University of California, Santa Barbara (UCSB)
Aug 2018 – June 2019

- Provided Tier 1 technical support for hardware, software, and network connectivity issues in a residential campus environment.
- Diagnosed and resolved Windows, macOS, Wi-Fi, Ethernet, TCP/IP, DNS, and DHCP issues.

Projects

Homelab Cloud + AI Platform - Self-Hosted Kubernetes/GitOps Infrastructure
Sep 2026 – Present

- Deployed a two-node Kubernetes (K3s) cluster (Raspberry Pi 5 control plane + ARM64 VM worker) with GitOps via Argo CD, GitHub Actions CI/CD behind a required PR gate, SOPS-encrypted secrets, Ansible, and Terraform (GitHub repo settings and branch rules)
- Found and fixed a production-style security gap where firewall rules never applied to Kubernetes traffic due to iptables chain ordering, verified with real blocked-vs-allowed traffic tests
- Root-caused network failures from packet-level evidence: a DNS resolver loop failing 21.6% of queries (fixed with authoritative reverse zones, to 0%) and cross-node pod traffic stuck in the router's flow cache (moved Flannel from a WireGuard tunnel to host-gw routing)
- Built a FastAPI backend (rate limiting, retries, metrics) with RAG over a local Ollama LLM and a chat assistant citing an offline 127 GB Wikipedia; evaluation-driven fixes doubled top-result accuracy (16 to 32 of 40)
- Built an iPhone calling bridge with the Pi as a Bluetooth hands-free unit (PipeWire HFP, D-Bus, FastAPI, WebSocket audio, RNNoise); used btmon captures to fix silent calls, then grew it into a Tailscale-reachable app with photo and file transfer
- Built a self-hosted webcam viewer: WebRTC streams straight from the camera host while a FastAPI gate handles invite-only passkey (WebAuthn) sign-in, hashed session tokens, sign-in throttling, and a strict CSP; shared via Tailscale Funnel
- Built a live touchscreen ops display (architecture map animated by real Prometheus request rates, one fish per pod sized by memory); it exposed an exporter bug inflating the DNS rate 200x (14,000 vs ~65/min), fixed with true counters
- Load tested with k6 (0% errors at ~470 req/s, cross-checked in Prometheus/Grafana); failure injection exposed a bug where a dead AI backend hung requests 180s+, fixed to fail cleanly in ~18s
- Automated encrypted backups with a recovery rehearsal that restores the control plane into a throwaway VM (same cluster CA, 23 pods running); email alerts later caught two silent failures, both root-caused and fixed

ParkingPulse - AI Vehicle Tracking & Data System
Aug 2024 – May 2025

- Engineered a real-time AI vehicle tracking platform using Python, OpenCV, and YOLOv8 for object detection and analytics
- Designed and implemented RESTful APIs to handle real-time data ingestion, processing, and retrieval
- Optimized system performance by reducing processing latency through frame skipping and model tuning

Undergraduate Research Assistant — UAV Data & Vision Systems
Aug 2024 – Mar 2025

- Developed OpenCV-based computer vision pipelines for real-time object tracking in UAV simulations
- Processed geospatial routing data using OSMnx to improve autonomous navigation efficiency
- Simulated autonomous systems in Webots to evaluate navigation performance and system reliability

Skills & Interests

- Languages: Python, JavaScript, TypeScript, C++, SQL
- Backend: Node.js, FastAPI, Flask, REST APIs
- Frontend: SvelteKit, HTML/CSS
- Cloud & DevOps: AWS (EC2, S3), GCP, Docker, Kubernetes (K3s), Argo CD, Ansible, Terraform, CI/CD
- Databases: PostgreSQL (pgvector), Redis, MongoDB, Firebase Firestore
- AI / Machine Learning: OpenCV, YOLOv8, FAISS, sentence-transformers, RAG, Ollama, Real-Time Data Processing
- Tools: Git, Linux, Prometheus, Grafana, k6, Tailscale, WebRTC, Bluetooth (btmon)

Certifications

- AWS Knowledge: Serverless
- Google Cloud: App Dev Environment
- Google Cloud: Load Balancing